Ex – 05	EDA – Data Inspection and Analysis

AIM

To inspect and analyze a dataset using Python Pandas by viewing DataFrames, filtering and subsetting data based on conditions, and calculating descriptive statistics such as mean, median, mode, range, variance, and standard deviation.

PROBLEM STATEMENT

To perform Exploratory Data Analysis (EDA) on a Student Performance dataset by examining its structure, selecting records using conditions, and computing measures of central tendency and dispersion to understand students' academic performance and the distribution of their marks.

PROCEDURE

1. Import the Pandas library for data inspection and analysis.
2. Load the Student Performance dataset from a CSV file.
3. Display the first and last five records using head() and tail().
4. Inspect the number of rows, columns, column names, and data types.
5. Check for missing values using isnull().sum().
6. Filter students whose math scores are greater than or equal to 80.
7. Select specific columns such as math score, reading score, and writing score.
8. Calculate the mean, median, and mode for the numerical columns.
9. Calculate the range, variance, and standard deviation of the numerical columns.
10. Display and interpret the results to understand the dataset's characteristics.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Step 1: Import Student Performance Dataset
df = pd.read_csv('Student_Perfomance.csv')
```

```
# Select subject columns
subjects = ['math score', 'reading score', 'writing score']
```

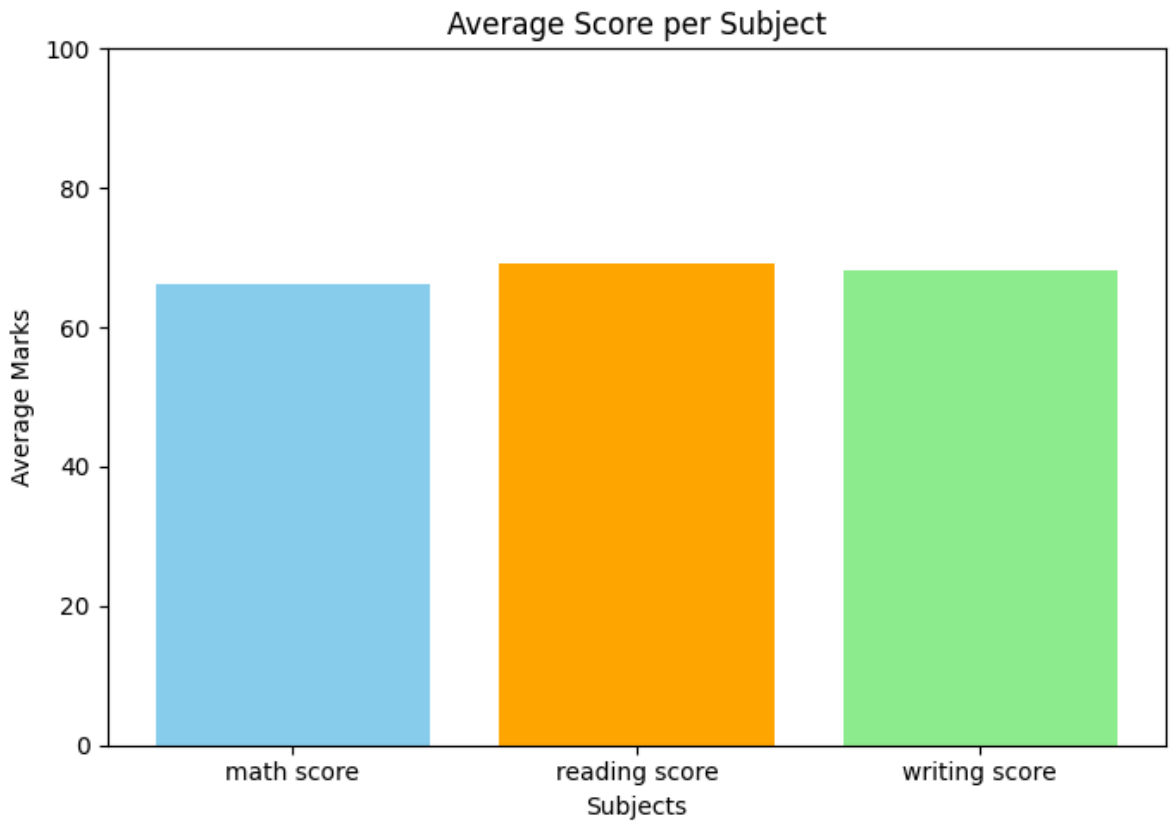
```
# Handle missing values
df[subjects] = df[subjects].apply(
    pd.to_numeric, errors='coerce'
)
df[subjects] = df[subjects].fillna(df[subjects].mean())
```

Average Score per Subject - Bar Plot

```
# Step 2: Average Score per Subject - Bar Plot
average_scores = df[subjects].mean()

plt.figure(figsize=(7, 5))
plt.bar(average_scores.index, average_scores.values,
        color=['skyblue', 'orange', 'lightgreen'])

plt.title('Average Score per Subject')
plt.xlabel('Subjects')
plt.ylabel('Average Marks')
plt.ylim(0, 100)
plt.tight_layout()
plt.show()
```

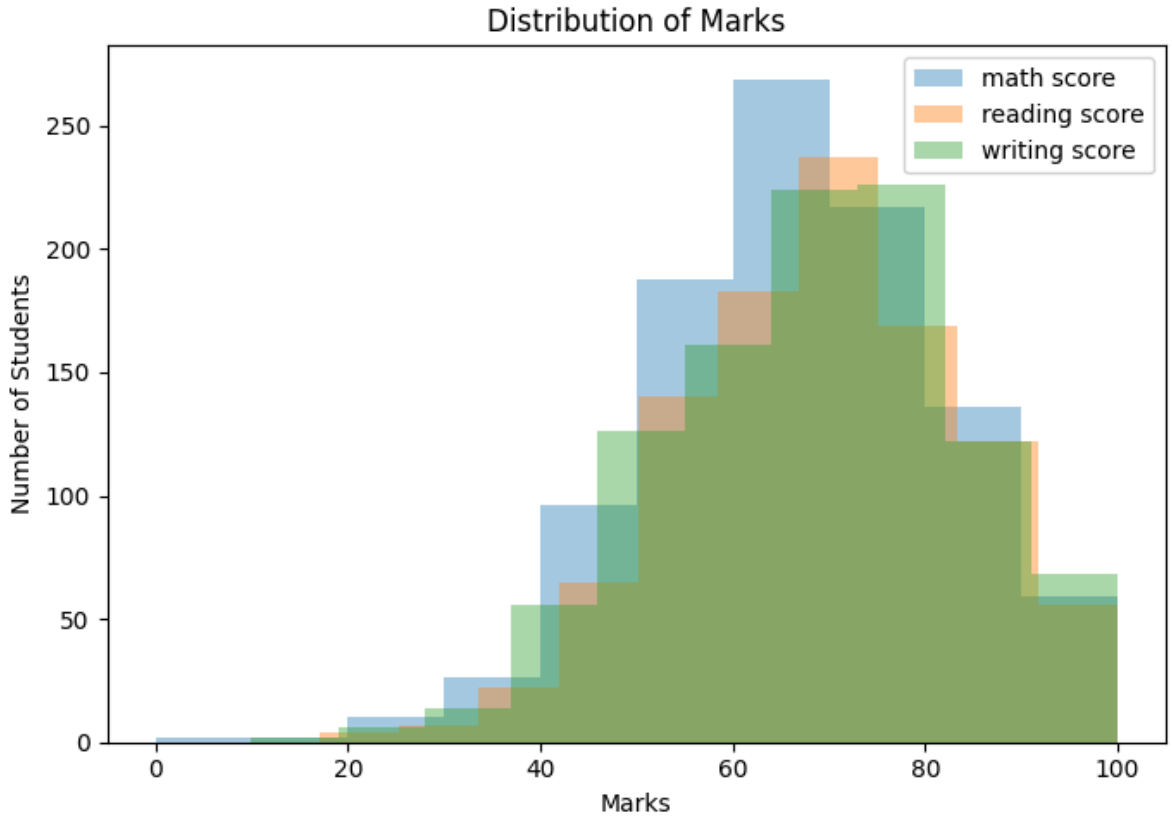


Distribution of Marks - Histogram

```
# Step 3: Distribution of Marks - Histogram
plt.figure(figsize=(7, 5))

for subject in subjects:
    plt.hist(df[subject], bins=10, alpha=0.4,
             label=subject)

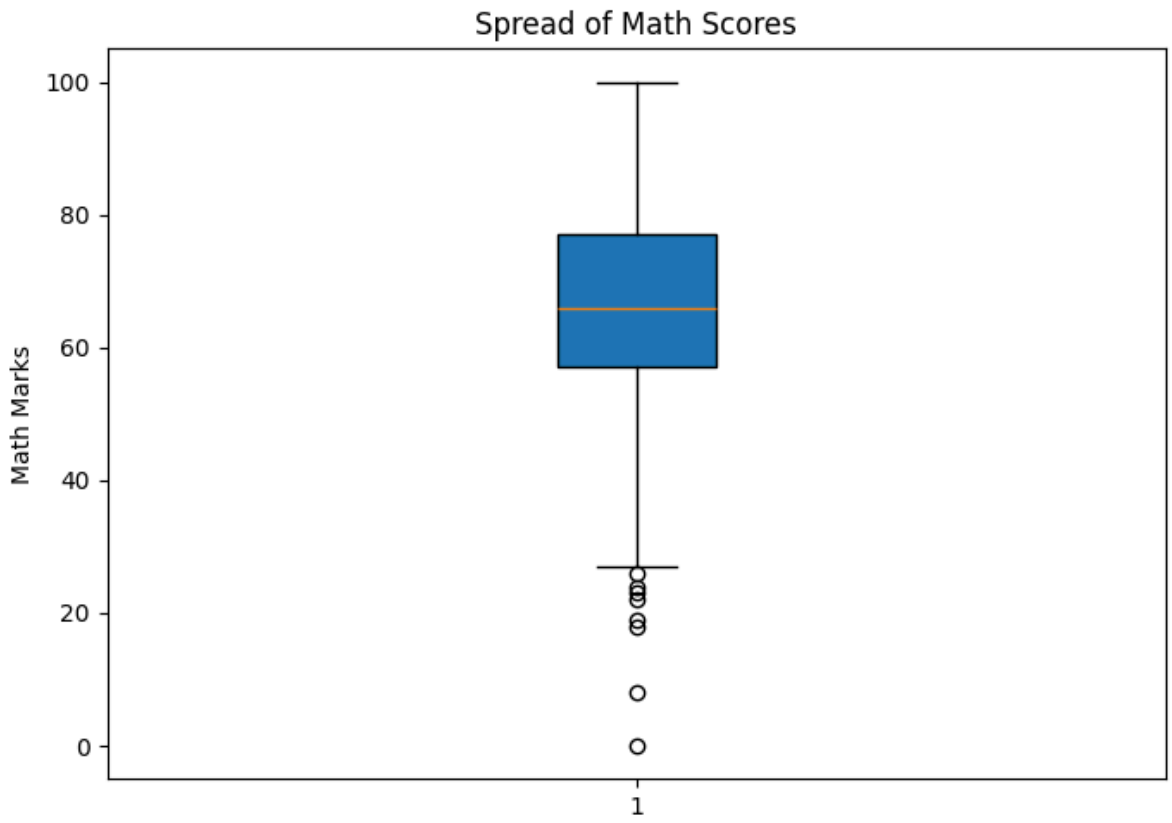
plt.title('Distribution of Marks')
plt.xlabel('Marks')
plt.ylabel('Number of Students')
plt.legend()
plt.tight_layout()
plt.show()
```



Spread of Math Score - Box Plot

```
# Step 4: Spread of Math Score - Box Plot
plt.figure(figsize=(7, 5))
plt.boxplot(df['math score'], patch_artist=True)

plt.title('Spread of Math Scores')
plt.ylabel('Math Marks')
plt.tight_layout()
plt.show()
```



Math Score Distribution with Mean,Median and Mode

```
# Step 5: Math Score Distribution with Mean,
# Median and Mode
math_scores = df['math score']

mean_value = math_scores.mean()
median_value = math_scores.median()
mode_value = math_scores.mode().iloc[0]

plt.figure(figsize=(8, 5))

plt.hist(math_scores, bins=10, color='skyblue',
         edgecolor='black', alpha=0.7)

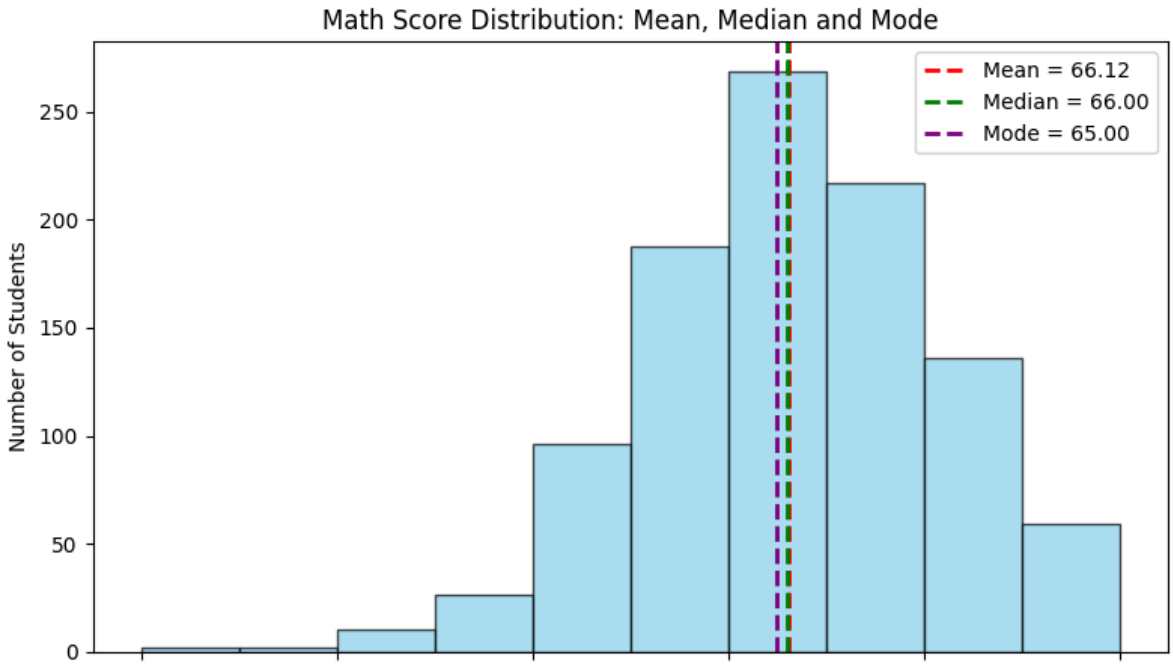
plt.axvline(mean_value, color='red', linestyle='--',
            linewidth=2, label=f'Mean = {mean_value:.2f}')

plt.axvline(median_value, color='green', linestyle='--',
            linewidth=2, label=f'Median = {median_value:.2f}')

plt.axvline(mode_value, color='purple', linestyle='--',
            linewidth=2, label=f'Mode = {mode_value:.2f}')

plt.title('Math Score Distribution: Mean, Median and Mode')
plt.xlabel('Math Score')
plt.ylabel('Number of Students')
plt.legend()
```

```
plt.tight_layout()
plt.show()
```



RESULT

The Student Performance dataset was successfully inspected and analyzed using Python Pandas. DataFrame structure, column names, data types, and missing values were examined, and records were filtered based on math scores.

The mean, median, mode, range, variance, and standard deviation were calculated for math, reading, and writing scores. These statistics help identify students' average performance, score distribution, and variability in academic achievement.